

The code is called from IDL as

```
res = call_external(libname, 'GET_MW', Lparms, Rparms, Sparms, Bparms, $  
Flux)
```

where libname is the library name (*.dll or *.so), and the parameters are:

0. Lparms – 4-element long integer (32-bit) array of dimensions and global integer flags.
1. Rparms – 4-element floating point (double, 64-bit) array of global real parameters.
2. Sparms – 5-element floating point (double, 64-bit) array of the source parameters.
3. Bparms – two-dimensional ($14 \times N_{\text{comp}}$) floating point (double, 64-bit) array describing the particle distributions.
4. Flux – two-dimensional ($3 \times N_f$) floating point (double, 64-bit) array receiving the computation results.

Namely, the above arrays have the following components:

Lparms = [Nf, Ncomp, Nnodes, FFflag]

where

0. Lparms[0] = Nf – number of frequencies in the spectrum.
1. Lparms[1] = Ncomp – number of considered plasma components.
2. Lparms[2] = Nnodes – number of energy nodes used for integration over energy in the continuous gyrosynchrotron code. Minimum value: 16; if Nnodes < 16, 16 nodes will be used instead.
3. Lparms[3] = FFflag – controls the free-free contribution: if zero, the free-free contribution (determined by the thermal plasma parameters n_{th} and T_0 , and computed using the approximate formulae from Dulk 1985) is included; if nonzero, the free-free contribution is ignored.

Rparms = [S, f_0 , Δf , μ_{acc}]

where

0. Rparms[0] = S – visible source area, in cm^2 .
1. Rparms[1] = f_0 – starting frequency of the spectrum, in Hz. If $f_0 > 0$, the emission frequencies are given by $f_0, f_0 \times 10^{\Delta f}, f_0 \times 10^{2\Delta f}, f_0 \times 10^{3\Delta f}, \dots$. If $f_0 \leq 0$, the emission frequencies are taken from the Flux[0, *] array.
2. Rparms[2] = Δf – logarithmic frequency step used to produce the spectrum (is used only when $f_0 > 0$).
3. μ_{acc} – the accuracy of determining the μ_0 parameter in the continuous algorithm. This parameter is used if $\mu_{\text{acc}} > 0$; otherwise the default value (10^{-3}) is used. Decreasing this parameter improves accuracy at the expense of the computation speed.

$$\text{Sparms} = [\Delta z, n_{\text{th}}, T_0, B, \theta]$$

where

0. Sparms[0] = Δz – source depth along line-of-sight, in cm.
1. Sparms[1] = n_{th} – thermal electron concentration, in cm^{-3} .
2. Sparms[2] = T_0 – thermal electron temperature, in K.
3. Sparms[3] = B – magnetic field strength, in G.
4. Sparms[4] = θ – viewing angle, in degrees.

Bparms = dblarr(14, Ncomp), where Ncomp is the number of the considered plasma components, and the parameters of the i -th component are specified as

0. Bparms[0, i] = q – particle charge, in units of the electron charge. E.g., $q = 1$ for electrons and muons, $q = -1$ for positrons and protons, $q = -2$ for α -particles.
1. Bparms[1, i] = m – particle mass, in units of the electron mass. E.g., $m = 1$ for electrons and positrons, $m \approx 1836$ for protons.
2. Bparms[2, i] = n – concentration of the considered particles, in cm^{-3} .
3. Bparms[3, i] = E_{id} – specifies the chosen energy distribution (non-integer parameters are rounded to the nearest integer).
4. Bparms[4, i] = either E_{min} or T :
 - a) E_{min} – the minimum particle energy for the power-law and double power-law distributions, in MeV;
 - b) T – the temperature for the thermal distribution, in K.
5. Bparms[5, i] = E_{max} – the maximum particle energy, in MeV.
6. Bparms[6, i] = E_{break} – the break energy in the double-power-law particle distributions, in MeV.
7. Bparms[7, i] = δ_1 – the low-energy power-law index.
8. Bparms[8, i] = δ_2 – the high-energy power-law index in the double-power-law particle distributions.
9. Bparms[9, i] = μ_{id} – specifies the chosen pitch-angle distribution (non-integer parameters are rounded to the nearest integer).
10. Bparms[10, i] = α_c or α_0 – either the loss-cone boundary in the loss-cone distributions or the beam direction in the beam-like distributions, in degrees.
11. Bparms[11, i] = $\Delta\mu$ – either the loss-cone boundary width or the beam angular width in the loss-cone or beam-like distributions, respectively.
12. Bparms[12, i] – *reserved*.
13. Bparms[13, i] = Nflag – specifies whether the considered particles make a contribution into the plasma dispersion: if nonzero, the concentration $n = \text{Bparms}[2, i]$ will be added to the thermal electron concentration $n_{\text{th}} = \text{Sparms}[1]$ before computing the plasma frequency (but will not affect the free-free emission); if zero, the particles are assumed to have no contribution into the dispersion.

Note that Ncomp can be zero; in such case, the Bparms parameter is not used and only the free-free emission is computed.

Flux = dblarr(3, Nf), where Nf is the number of the emission frequencies. After calling the function, this array contains the computation results (and can specify the emission frequencies on input), namely:

0. Flux[0, *] – emission frequencies, in GHz. On input, if Rparms[1] = $f_0 \leq 0$, the frequencies are taken from this array; otherwise, the frequencies are computed automatically.

1. Flux[1, *] – X-mode emission intensities, in sfu.

2. Flux[2, *] – O-mode emission intensities, in sfu.

Returned value: 0 if there were no errors; -1 if there was an insufficient number of parameters (< 5).

Currently, the following energy distributions are implemented:

- Thermal distribution (THM, Eid = 2).
- Power-law over the particle energy (PLW, Eid = 3).
- Double power-law over the particle energy (DPL, Eid = 4).
- Power-law over the particle flux (PLF, Eid = 13).
- Double power-law over the particle flux (DPF, Eid = 14).

The following pitch-angle distributions are implemented:

- Isotropic (ISO, $\mu_{id} = 0$ or 1).
- Symmetric gaussian loss-cone (GAU, $\mu_{id} = 3$).
- Gaussian beam (GAB, $\mu_{id} = 4$).

If either Eid or μ_{id} do not match the above values, no emission is computed for the corresponding particle component (although the concentration may be added to the thermal electron concentration, if Nflag $\neq 0$).